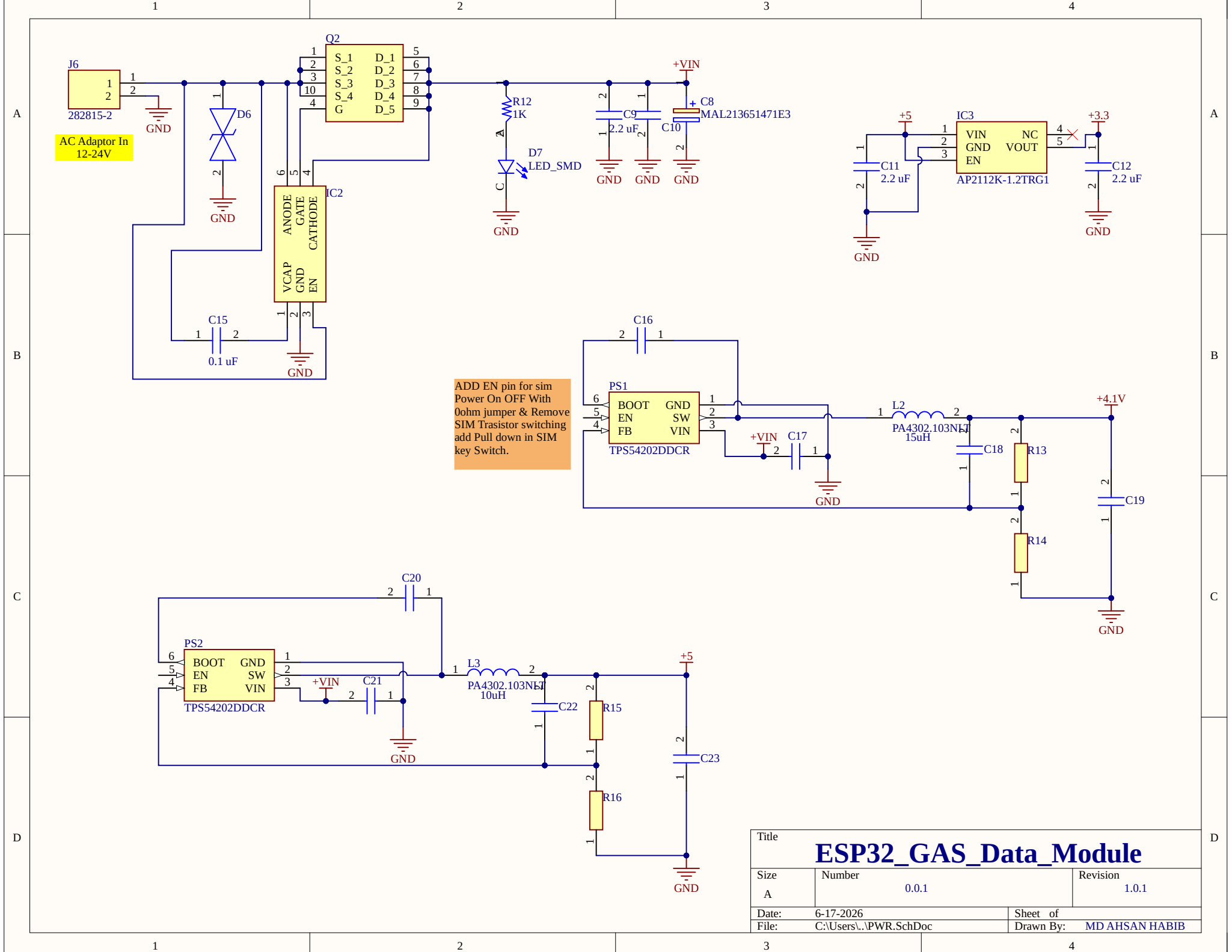
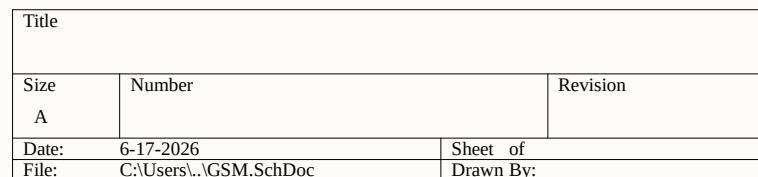


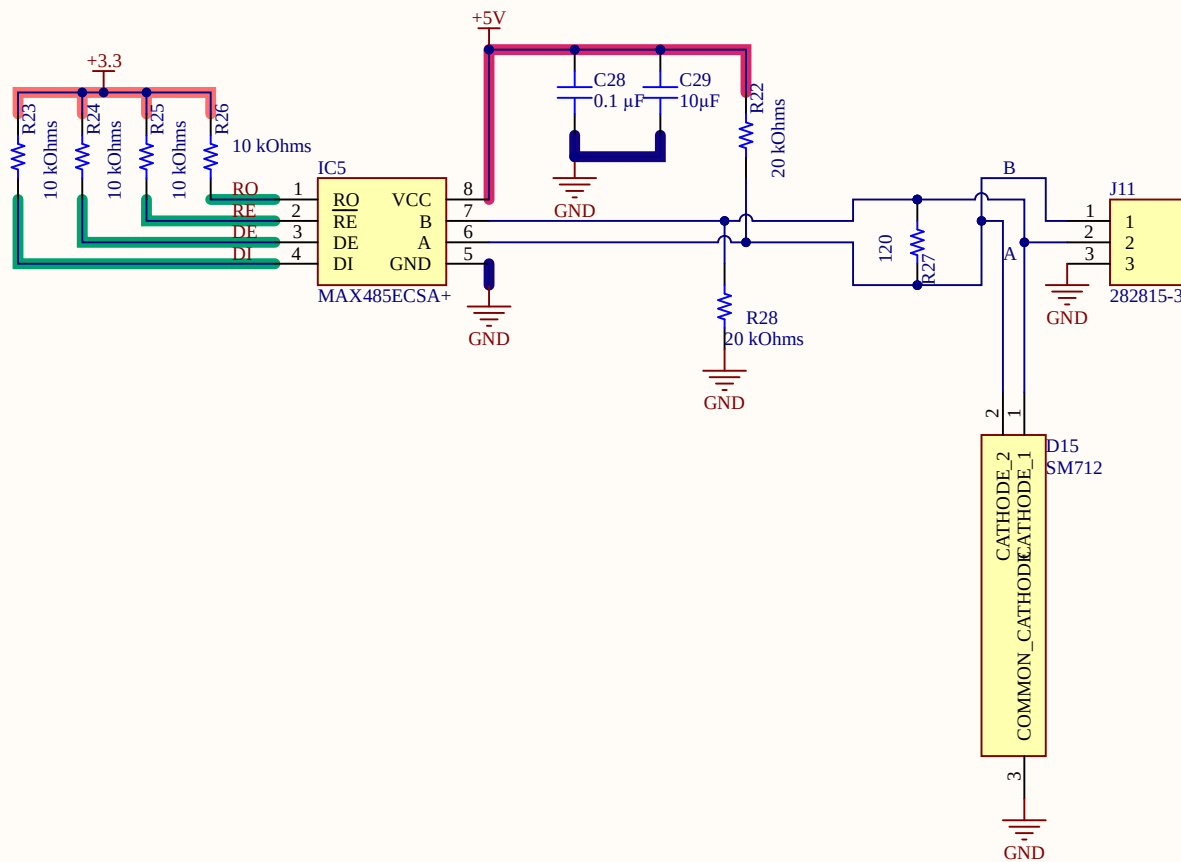
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[LCSC link](#)

Pin	IO	Function
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3	3	3V3
4	4	EN
5	5	SENSOR_VP
6	6	SENSOR_VN
7	7	IO34
8	8	IO35
9	9	IO32
10	10	IO33
11	11	IO25
12	12	IO26
13	13	IO27
14	14	IO12
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18	18	GND_6
19	19	GND_7
20	20	GND_8
21	21	GND_9
22	22	GND_10
23	23	GND_11
24	24	GND_12
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26	26	IO4
27	27	IO16
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31	31	IO19
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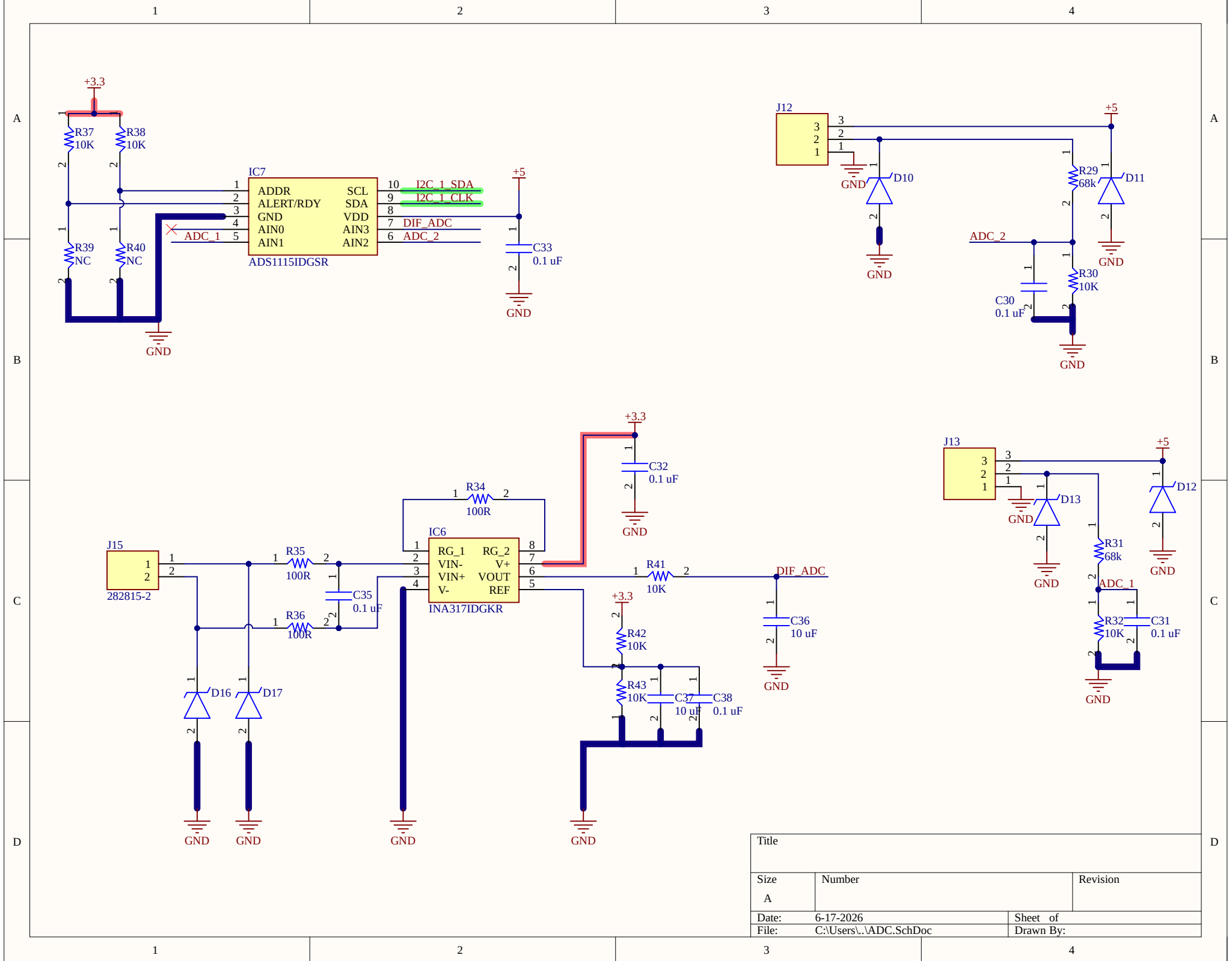


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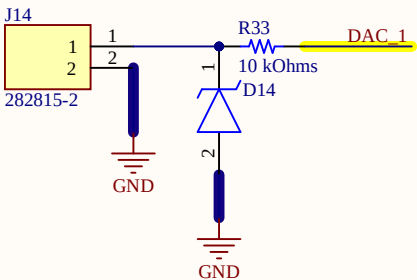




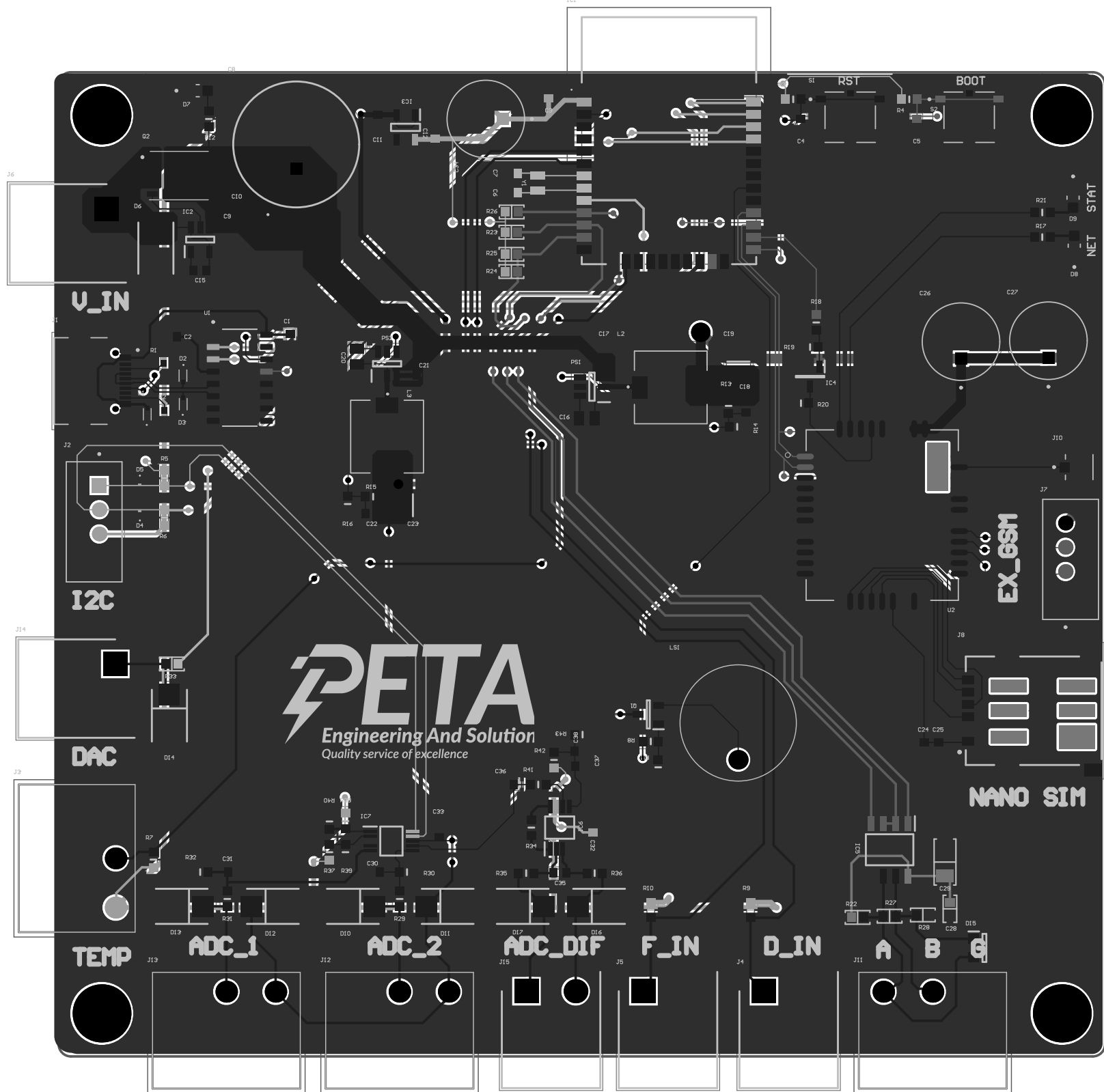
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Comment	Description	Designator	Footprint	LibRef	Quantity
0.1 uF	Capacitor	C1, C2, C3, C4, C5, C15, C30, C31, C32, C33, C35, C38	0408003	LC0603040R7888B104	12
1.5 uF	Capacitor	C8, C7	0408003	LC0603040R7888B104	2
ML2136S147J3	Capacitor Polarised	C9	0408003000M0000100H2700	ML2136S147J3	1
2.2 uF	Capacitor	C9, C11, C12, C34, C35	0408003	LC0603040R7888B104	5
C5750SRH106R230K A	Capacitor	C10	C5750_Commercial	C5750SRH106R230K A	1
C385C10M59PC7U	Capacitor	C36, C20	0408003000M0000100H2700	C385C10M59PC7U	2
C5750SRH106R230K A	Capacitor	C17, C21	0408003	C5750SRH106R230K A	2
GM035C1E560A01 D	Capacitor	C18	0408003	GM035C1E560A01 D	1
GM035R1C476R3E5 L	Capacitor	C19, C23	0408003000M0000100H2700	GM035R1C476R3E5 L	2
CLALAC10G1880L100 9A	Capacitor	C22	0408003	CLALAC10G1880L100 9A	1
470 uF	220F 25V Aluminum Electrolytic Capacitors Radial, Dia 20.00 mm (0.787 in)	C26, C27, C34	0408003000M0000100H2700	ELLFC12221	3
0.1 uF	0408003 0.1uF 50V X7R0003	C28	0408003_1068	C385C10M59PC7U11	1
10 uF	0408003 0.1uF 50V X5R1005	C29	0408003_1068_3525-1.8MM	LC0603040R7888B104	1
10 uF	Capacitor	C36, C37	0408003	LC0603040R7888B104	2
LES06D5.0CT1G	TVS Diode (Bi-directional)	D1, D2, D8	LES06D5.0CT1G	LES06D5.0CT1G	3
DES03V3218C3F-7	TVS Diode (Bi-directional)	D4, D5	DES03V3218C3F-7	DES03V3218C3F-7	2
9MLB24CA-13-F	TVS Diode (Bi-directional)	D6	DIONM5336C250N	9MLB24CA-13-F	1
LED_SMD	Red 635nm LED Indication - Discrete 2.1x0.9mm (2032 Metric)	D7, D8, D9	LEDXCL2500090-2N	8990110007F	3
9MLB5.0A	TVS Diode (Uni-directional)	D10, D11, D12, D13, D14, D16, D17	9MLB5.0A	9MLB5.0A	7
9M712	TVS Diode (Bi-directional)	D15	SDT06P240X15-3N	9M712	1
ESP32-WROOM-32E-N8	Integrated Circuit	IC1	ESP32-WROOM-32E-N8	ESP32-WROOM-32E-N8	1
LM74700QDBVTQ1	Integrated Circuit	IC2	SDT06P280X145-6N	LM74700-Q1M0K-1_NCPB	1
AP2112K-CL2TRFCL	Integrated Circuit	IC3	SDT06P280X145-6N	AP2112K-CL2TRFCL	1
MMBT222AL1TG	Integrated Circuit	IC4	SDT06P237X111-3N	MMBT222AL1TG	1
MAX485ECS+	Integrated Circuit	IC5	SDCL27P600X175-8N	MAX485ECS+	1
INA317DCGR	Integrated Circuit	IC6	SDT06P480X110-8N	INA317DCGR	1
AD8115DCGR	Integrated Circuit	IC7	SDT06P480X110-10N	AD8115DCGR	1
TYPEC-31-M-12	1.81 Connectors 38 Receptacle 1 8.94 7.3mm R04E5	J1	HPD_TYPEC-31-M-12	TYPEC-31-M-12	1
DISPLAY	Connector Header Through Hole 4 position 0.098 (2.50mm)	J2	JST_B4B-XH-A(F)(SN)	B4B-XH-A(F)(SN)	1
DSB000 PT100	Connector	J3	2828153	282815-3	1
Digital	Connector	J4	SDT06P240X15-3N	282815-2	1
FLOAT_SNS	Connector	J5	SDT06P240X15-3N	282815-2	1
282815-2	Connector	J6, J4, J5	SDT06P240X15-3N	282815-2	3
0X_3M	Connector Header Through Hole 4 position 0.098 (2.50mm)	J7	JST_B4B-XH-A(F)(SN)	B4B-XH-A(F)(SN)	1
9M000-6-1-14-00-A	6 Position Card Connector VARIO 3M Surface Mount, Right Angle Gold	J8	GCT_9M000-6-1-14-00-A	9M000-6-1-14-00-A	1
SMD_SMD	Connector Header Through Hole 4 position 0.098 (2.50mm)	J9	SMD_SMD	SMD_SMD	1
CONJL001-SMD	Connector	J10	CONJL001-SMD	CONJL001-SMD	1
282815-3	Connector	J11, J12, J13	2828153	282815-3	3
PAW302 103NLT	Inductor	L2, L3	PAW302103NLT	PAW302 103NLT	2
OM-1259C-0585T	Inductor Speaker or Buzzer	L5L	OM1259C0585T	OM-1259C-0585T	1
TPS64020DCR	Power Supply	PS1, PS2	SDT06P280X110-6N	TPS64020DCR	2
CMG315U17	Integrated Circuit	Q1	SDT06P240X110-3N	CMG315U17	1
REC101P04015A01MA1	AC/DC (N-Channel)	Q2	REC101P04015A01MA1	REC101P04015A01MA1	1
CP040402100RFEH4 P	Resistor	R1, R2	RES0603_100N	CP040402100RFEH4 P	2
10K	Resistor	R3, R4, R11, R10, R12, R17, R18, R19, R12, R13	RES0603	RES	10
4.7K	Resistor	R5, R6, R7, R8, R10, R18	RES0603	RES	6
100K	Resistor	R9	RES0603	RES	1
1K	Resistor	R12, R17, R21	RES0603	RES	3
CP040402100RFEH4 P	Resistor	R13, R15	RES0603	CP040402100RFEH4 P	2
CP040402193RFEH4 P	Resistor	R14	RES0603	CP040402193RFEH4 P	1
CP040402130RFEH4 P	Resistor	R16	RES0603	CP040402130RFEH4 P	1
1K	Resistor	R19	RES0603	RES	1
NC	Resistor	R10, R19, R10	RES0603	RES	3
20 kOhms	RES SMD 10K OHM 1% 1/10W 0603	R22, R28	RES0603_100H	RT0603PF07104L	2
10 kOhms	RES SMD 10K OHM 1% 1/10W 0603	R23, R24, R25, R26, R27	RES0603_100H	RT0603PF07104L	5
120	RES SMD 10K OHM 1% 1/10W 0603	R27	RES0603_100H	RT0603PF07104L	1
56K	Resistor	R29, R31	RES0603	RES	2
100K	Resistor	R34, R35, R36	RES0603	RES	3
PT3206_5MG15_SMT8 Z_LFS	Resistor	SL, S2	SW_PT3206_5MG15_SMT8_Z_LFS	PT3206_5MG15_SMT8_Z_LFS	2
0-G40C	USB to serial chip	U1	SDCL27P600X175-8N	0-G40C	1
9M800C	Module GSM/Bluetooth; GPS; 2G, 1800MHz; 300MHz; 85.9kbits; SMD	U2	LC1110P1570X1780X90-42N	9M800C	1
CMV-T1A-32-76894-6-PPF-20PM-TA-QC	Tuning Fork Crystal 32.768kHz in SMD 1.2 x 2.0mm package	Y1	XTAL_CMV-T1A-32-76894-6-PPF-20PM-TA-QC	CMV-T1A-32-76894-6-PPF-20PM-TA-QC	1